II-Probabilistic Approaches Likelihoods, empirical distributions and learning

Foundations of Neuro-Symbolic AI

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Outline

- Log-likelihood
- Learning Bayesian networks
- Empirical Distributions

How to select a model given data?

Let us consider the typical learning problem:

- The distribution $\mathbb{P}\left[X_{[d]}\right]$ of variables $X_{[d]}$ is unknown
- We observe m samples

  $$x^j_{[d]} = (x^j_0, \ldots, x^j_{d-1})$$

  enumerated by $j \in [m]$.

We assume that each sample $x^j_{[d]}$ is drawn

- independently from each other and
- identically distributed by a distribution $\mathbb{P}\left[X_{[d]}\right]$.

This assumption is abbreviated by i.i.d..

The probability of drawing a dataset $X^{[m]}_{[d]}$ from a distribution $\mathbb{P}\left[X_{[d]}\right]$ is then

$$\prod_{j \in [m]} \mathbb{P}\left[X_{[d]} = x^j_{[d]}\right] .$$

This is the likelihood of $\mathbb{P}\left[X_{[d]}\right]$.

The likelihood principle

We make the hypothesis that the distribution is from a set Γ of possible distributions. A natural choice would be to choose the distribution maximizing the likelihood

$$\operatorname{argmax}_{\mathbb{P}[X_{[d]}]\in\Gamma} \left(\prod_{j\in[m]} \mathbb{P}\left[X_{[d]} = x^j_{[d]}\right] \right) .$$

Sometimes we regularize, that is prefer simple distributions over more complex ones.

Likelihood principle

The selection of a distribution should depend on the data $X^{[m]}_{[d]}$ only through the likelihood

$$\prod_{j\in[m]} \mathbb{P}\left[X_{[d]} = x^j_{[d]}\right] .$$

Negative log-likelihood

We simplifiy the likelihood by taking the log and averaging. This is the log-likelihood

$$\frac{1}{m} \ln \left[\prod_{j \in [m]} \mathbb{P}\left[x^j_{[d]} \right] \right] = \frac{1}{m} \sum_{j \in [m]} \ln \left[\mathbb{P}\left[x^j_{[d]} \right] \right] .$$

The maximum likelihood method corresponds then with the minimization of the negative log-likelihood (NLL)

$$\operatorname{argmin}_{\mathbb{P}[X_{[d]}] \in \Gamma} \sum_{j \in [m]} - \ln \left[\mathbb{P}\left[X_{[d]} = x^j_{[d]} \right] \right] .$$

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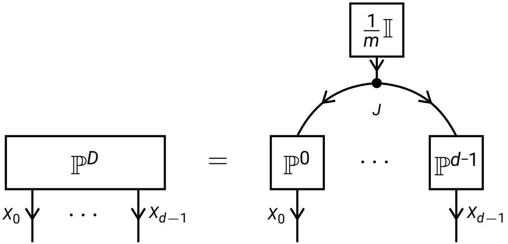
Empirical distributions

Let us understand the uniform choice of a sample from $x_{[d]}^{[m]}$ as a distribution

$$\mathbb{P}^D\left[X_{[d]}\right] = \frac{1}{m}\sum_{j\in[m]} \epsilon_{x^j_{[d]}}\left[X_{[d]}\right]$$

which we call empirical distribution.

We have the CP decomposition of the empirical distribution



where

$$\mathbb{P}^k\left[X_k|J\right] = \sum_{j\in[m]} \epsilon_{x^j_k}\left[X_k\right] \otimes \epsilon_j\left[J\right] .$$

Perspective of basis calculus

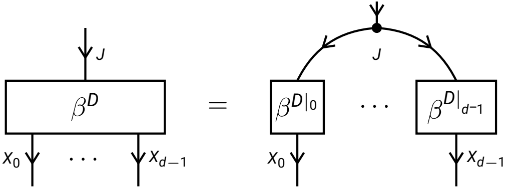
We understand the empirical distribution to samples $x_{[d]}^{[m]}$ as the basis encoding of the sample selecting map

$$D\ :\ [m] \to \bigtimes_{k\in[d]} [m_k]$$

defined for $j \in [m]$ by $D(j) = x_{[d]}^{j}$. To be more precise we have

$$\mathbb{P}^{D}\left[X_{[d]}\right] = \left\langle \beta^{D}\left[X_{[d]}, J\right], \frac{1}{m}\mathbb{I}\left[J\right] \right\rangle_{\left[X_{[d]}\right]} .$$

The CP decomposition of the empirical distribution is then based on the decomposition of the basis encoding into the basis encodings of its coordinate restrictions:



Cross-entropy

The cross-entropy between two distributions $\tilde{\mathbb{P}}[X_{[d]}]$ and $\mathbb{P}\left[X_{[d]}\right]$ is

$$\mathbb{H}\left[\tilde{\mathbb{P}}, \mathbb{P}\right] = \left\langle \tilde{\mathbb{P}}[X_{[d]}], -\ln\left[\mathbb{P}\left[X_{[d]}\right]\right] \right\rangle_{[\varnothing]} .$$

The negative log-likelihood is the cross-entropy between the empirical distribution and the candidate $\mathbb{P}\left[X_{[d]}\right]$.

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Two learning tasks

- Structure learning: Find the hypergraph $\mathcal{G}$
- Parameter estimation: Find the best decorations by conditional probability distributions

Decomposing the negative log-likelihood

For a Bayesian network we have for any index tuple $x_{[d]}$

$$\ln\left[\mathbb{P}\left[X_{[d]} = x_{[d]}\right]\right] = \sum_{k\in[d]} \ln\left[\mathbb{P}\left[X_k = x_k \middle| X_{\mathrm{Pa}(k)} = x_{\mathrm{Pa}(k)}\right]\right] .$$

The NLL thus decomposes into local terms

$$\begin{aligned}
&\left\langle \mathbb{P}^D\left[X_{[d]}\right], -\ln\left[\mathbb{P}\left[X_{[d]}\right]\right]\right\rangle_{[\varnothing]} \\
&= \left\langle \mathbb{P}^D\left[X_{[d]}\right], \sum_{k\in[d]} -\ln\left[\mathbb{P}\left[X_k \middle| X_{\mathrm{Pa}(k)}\right]\right]\right\rangle_{[\varnothing]} \\
&= \sum_{k\in[d]} \left(\left\langle \mathbb{P}^D\left[X_{[d]}\right], -\ln\left[\mathbb{P}\left[X_k = x_k \middle| X_{\mathrm{Pa}(k)} = x_{\mathrm{Pa}(k)}\right]\right]\right\rangle_{[\varnothing]}\right)
\end{aligned}$$

The minimum can found separately for each k by the minimization problem

$$\mathrm{argmin}_{\mathbb{P}\left[X_k \middle| X_{\mathrm{Pa}(k)}\right]} \frac{1}{m} \sum_{j\in[m]} -\ln\left[\mathbb{P}\left[X_k = x_k \middle| X_{\mathrm{Pa}(k)} = x_{\mathrm{Pa}(k)}\right]\right] .$$

Optimum by normalization of the empirical distribution

The optima for each conditional distribution is taken at

$$\mathrm{argmin}_{\mathbb{P}\left[X_k \middle| X_{\mathrm{Pa}(k)}\right]} \left\langle \mathbb{P}^D, -\ln\left[\mathbb{P}\left[X_k = x_k \middle| X_{\mathrm{Pa}(k)} = x_{\mathrm{Pa}(k)}\right]\right] \right\rangle_{[\varnothing]}$$
$$= \left\langle \mathbb{P}^D \left[X_{[d]}\right] \right\rangle_{\left[X_k | X_{\mathrm{Pa}(k)}\right]}$$

This can be verified by the first order criterion (vanishing gradients).

Parameter estimation

Parameter estimation for Bayesian networks

- The likelihood of Bayesian networks decomposes into local terms.
- Best log-likelihood maximizers are the conditional probability distributions.